

Prospects & Overviews

Cancer adaptations: Atavism, de novo selection, or something in between?

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From an evolutionary perspective, both atavism and somatic evolution/convergent evolution theories can account for the consistent occurrence, and astounding attributes of cancers: being able to evolve from a single cell to a complex organized system, and malignant transformations showing significant similarities across organs, individuals, and species. Here, we first provide an overview of these two hypotheses, including the possibility of them not being mutually exclusive, but rather potentially representing the two extremes of a continuum in which the diversity of cancers can emerge. In reviewing the current literature, we also discuss the criteria that should be applied to discriminate between the two competing theories and to determine their relevant contributions to oncogenesis and cancer progression. Finally, we deliberate on the potential applications of this conceptual framework in developing novel treatment strategies.

Keywords:

atavism; cancer; comparative oncology; convergent evolution; somatic selection

Introduction

Cancer is a disease of multicellular organisms that occurs throughout the animal kingdom, from hydra to whales [1]. An

intriguing aspect of this pathology is that, in the space of only few months or years, it can evolve from a single cell to a highly complex and well organized system – i.e. the solid tumor – that is comparable to a functional, though abnormal, organ (see [2] for a synthesis). Another intriguing aspect of this disease is that although **each cancer evolves as a novel entity** (except for transmissible cancers, see [3, 4]), there are significant similarities between the attributes of malignant cells originating from different organs, and from different individuals and species; furthermore, there are large similarities across the diversity of cancers in the way cancer develops and progresses inside the host's body (e.g. anarchic proliferation, genomic instability, immortality, insult resilience, aerobic glycolysis, invasive metastatic cascade, and host destruction). **From an evolutionary perspective, two main hypotheses can account for the consistent occurrence of these idiosyncratic traits. The first proposes that cancer results from atavism, the second involves somatic evolution and convergent evolution.** Distinguishing between the two hypotheses is not a semantic problem, but a necessity, because atavistic programs could represent a novel therapeutic target, whereas stochastic passive mutability does not (i.e. different therapeutic implications).

The atavistic model

In the atavistic model, for instance as proposed by Davies and Lineweaver [5], **cancer is interpreted as a reversion to phylogenetically prior capabilities, namely the release of a highly conserved survival program encrypted in every eukaryotic cell, and hence in every multicellular animal.** This ancient program would have evolved during the pre-Cambrian period, when the selective forces acting on unicellular organisms were favoring adaptations prioritizing the continuation/proliferation of cellular life in different, often adverse, environmental conditions. Because metazoans evolved from unicellular ancestors, strong constraints on preventing inappropriate atavisms (e.g. uncontrolled cell proliferation) must have evolved concomitantly to build complex and functional multicellular organisms [6]. These adaptations, however, did not necessarily eliminate ancestral programs, but rather prevented their expression, at least during the

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period of organismal lifetime when natural selection is most intense (i.e. the first part of life). When the control fails, the atavism paradigm holds that a competitive struggle between the battle-scarred protozoan (pre-Cambrian) phenotype and the “normal” metazoan phenotype ensues, and cancer emerges [7]. The atavistic model is a plausible and elegant explanation of cancer for several reasons. First, it is chronologically relevant because multicellularity derives from unicellularity (see [8, 9]). It also provides a coherent explanation for the commonality of traits displayed by malignant cells from different species (i.e. ancestral legacy). Additionally, the atavistic theory remarkably explains some of the distinct ecological niches favored by cancer cells, for instance the ability of malignant cells to flourish and proliferate in anoxic/hypoxic conditions (which prevailed at the transition to multicellularity). Finally, because the “ancient phenotype” has actually evolved over eons to overcome the effect of stressors, it provides an evolutionary explanation for the difficulties of treating and curing cancer [10] – radiation and chemotherapeutic drugs being examples of “stressors” in this context.

The somatic selection model

Contrary to the atavistic model, the somatic selection hypothesis holds that cancer adaptations (whatever their proximate causes, e.g. genetic, epigenetic, adaptive phenotypic plasticity...) are under strong selection each time a novel malignancy occurs. Each primary tumor originates from a single cell that has to “reinvent the wheel” given that its evolved, malignant, products die with the host [11], but see the case of transmissible cancers [4]. The highly sophisticated level of organization that coalescent cancer cells achieve in few months or years in tumors within an individual, and laterally between individuals and across species, would be an illustration of convergent evolution – somatic selection operating in environments that are broadly governed by the same ecological constraints (i.e. the body of multicellular organisms). Somatic selection is particularly efficient because extensive and heritable variation (including genetic, cytogenetic, epigenetic, and phenotypic diversity) exist within tumors (i.e. intratumor heterogeneity, hereafter “ITH”) [12].

Lloyd et al. [13] proposed that at least some ITH in the molecular properties of cancer cells are governed by predictable regional variations in environmental selection forces. Thomas et al. [14] recently hypothesized that ITH could itself be the result of a selective process, bet-hedging, that starts from the very first steps of oncogenesis. In evolutionary ecology, bet-hedging is defined as a strategy that reduces the temporal variance in an organism’s fitness at the expense of a lowered arithmetic mean fitness. Thomas et al. [14] adapted this definition to oncogenesis by considering it to be a process that reduces the temporal variance in the success of a cancerous or precancerous cell relative to that of other cells in the organism. “Successful” tumors would be those favored by somatic selection early in oncogenesis, and hence these cells that will be selected to generate variation in their progeny in order to produce the diversity of cell components vital to build, de novo, an intricate cooperative system like solid

tumors (i.e. bet-hedging would be a de novo trait favored by oncogenic selection). Akin to the atavistic model, this scenario can account for several cancer characteristics. For example, with an evolutionary persistence of maximally only a few decades, bet-hedging is expected to be poorly tuned, hence yielding numerous aberrant/non functional cells (the case in tumors), instead of producing different but functional entities, as has been observed in organisms employing bet-hedging strategies for eons (e.g. plants in deserts) [14]. Also, even though bet-hedging could facilitate de novo development of tumor complexity, it is expected to be a slow process given that only a few useful components are produced per time unit; therefore, potentially explaining the late occurrence of solid tumors in a lifetime.

Between atavism and de novo selection, where is the truth?

The two hypotheses are, in fact, not mutually exclusive. For instance, **the re-emergence of an ancestral program can well be followed by selective processes, provided substantial variation is generated among malignant cells.** Resistance to therapies convincingly demonstrates the existence of selective processes. The true question is thus rather to establish the relative contribution of the two evolutionary processes to cancer development and progression. In an extreme scenario of the atavistic model, it could be argued that the minimum program that needs to be reactivated for cancer initiation is the one leading to bet-hedging (see for instance [15]); from this, somatic selection alone can progressively yield more and more sophisticated malignant adaptations. That is, the ancestral program encrypted in this extreme case would only preserve the cells’ evolvability rather than specific adaptations. This is possible, but there is in fact no need to invoke an ancestral program to explain the induction of an early bet-hedging during the oncogenesis: random mutations during the organism’s life can potentially induce it too. At present it appears challenging to determine which hypothesis (or their relative contribution, if both occur) is the more parsimonious and the more likely to explain cancer, because both seem to be able to account for the many characteristics of malignancies. **Even if there is no strict need to invoke atavism to explain cancer (i.e. selection is enough), atavism can nevertheless remain the underlying cause of cancer originating from the evolutionary history of life** (i.e. multicellularity derives from unicellularity). It is also possible that different cancers rely on the two evolutionary trajectories, i.e. certain cancers could mostly be governed by atavisms, while others could progress following random mutations that generate bet-hedging followed by somatic selection.

Criteria to discriminate between competing theories

Discriminating between the two theories basically requires the development of methodologies that would resolve whether cancer adaptations, along tumorigenesis, result from the

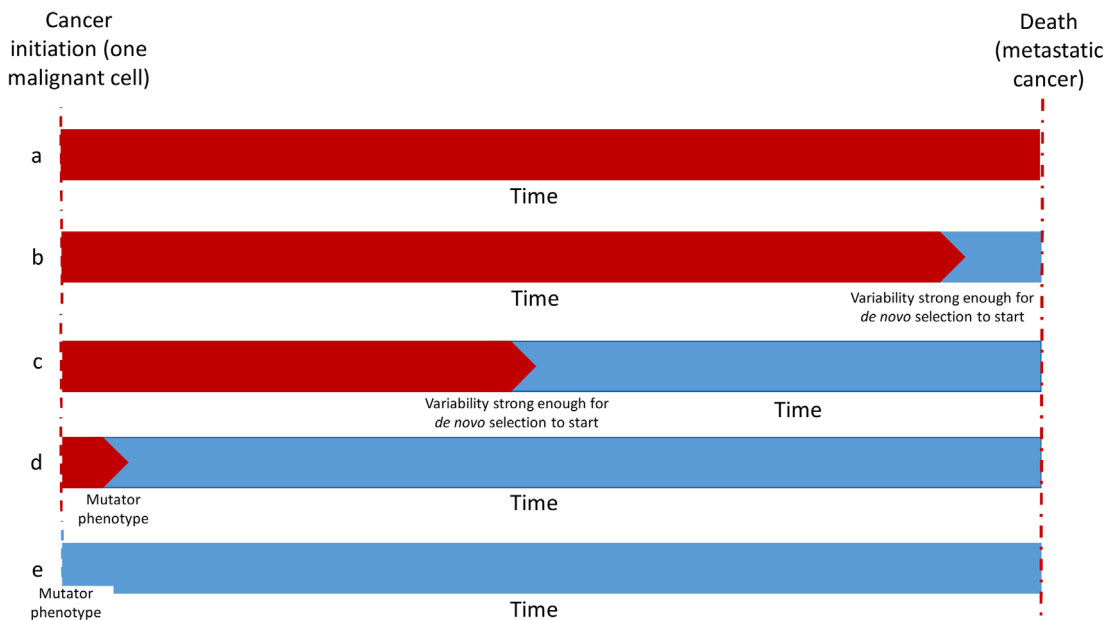


Figure 1. Different scenarios depicting the relative importance of atavism and de novo selection during tumorigenesis. **A:** Cancer is only due to the reactivation of ancestral programs. **B:** Cancer is mostly due to the reactivation of ancestral programs, and de novo selection occurs only at later stages of the tumorigenesis. **C:** Both atavism and de novo selection mutations contribute to tumorigenesis. **D:** Atavism is relevant only at the beginning of the tumorigenesis (e.g. early reactivation of an ancestral program allowing bet-hedging). **E:** Cancer is only due to de novo selection, without the need to invoke atavism.

reactivation of ancient programs, or conversely from novel mutations coupled with de novo selection. As illustrated in Fig. 1, extreme scenarios (cases A and E) are that cancer can be explained entirely by atavism (even drug resistance in final metastatic stages, e.g. [10, 15]), or exclusively by novel mutations (originating from a mutator phenotype from onset) followed by de novo selection. If both theories are valid, the challenge for scientists is to determine when the shift in their relative importance occurs. Teasing apart the evolutionary origins of malignant transformations is further complicated by the two potential origins of bet-hedging. Bet-hedging can be de novo selected, or as previously described, it can be the encoded minimal ancestral program. Bet-hedging being a common, most likely ancient, organismal trait [15, 16], is therefore compatible with atavism. Thus, distinguishing between the two hypotheses, atavism and somatic mutations, particularly in light of bet-hedging, can prove challenging. With this in mind, we can propose several research directions that would be useful to explore.

In a first step, we suggest looking for ancestral character states (genetic, epigenetic, and regulatory markers) that are homologous in unicellular and multicellular organisms, and are also activated in malignant cells and hence contribute to cancer development, e.g. cell death, stress responses, DNA repair mechanisms, etc. [15]. **Monophyletic characters will indicate that the history of these characters involves a single origin, thereby supporting the atavism theory. For example, the altruistic cell death mechanism observed in bacteria in monophyletic group, or clade, is a group of organisms that consists of all the descendants of a common ancestor**

response to antibiotics has also been documented in cancer cells that have been exposed to radiotherapy [17, 18], potentially indicating the ancient origins of these processes. Determining whether these character states are homologous or are the results of convergent evolution requires in depth phylogenetic/phylogenomic analyses. Along these lines, a research program would consist of exploring whether in the genome of protozoans we can detect areas that have strong similarities with those that are specifically activated in metazoan malignant cells along the tumorigenesis process. If this is the case, we then suggest exploring the biological function of these areas in protozoans. The atavistic model predicts that there should be such similarities, and that shared areas are involved in protozoan adaptive functions that have an “oncogenic value” when transposed in a cancer metazoan context (e.g. proliferation, bet hedging strategy, resistance to adverse environmental conditions ...). The de novo selection model conversely predicts that if any similarities exist, there would be no correspondence between their function in protozoan organisms and the metazoan malignant process. We reciprocally suggest exploring the molecular determinants of bet-hedging in several microorganisms that use this strategy routinely, and then comparing them to those acting in the case of malignant cells. Determining whether the mutation rate in cancer cells changes as a function of the environmental stress would be crucial to explore because if it did, the mechanism causing it would start to look more like a programmatic adaptation rather than just a random series of errors (i.e. the SOS regulon program in bacteria is a mutator program that can be upregulated under environmental stress [19]). In addition to bet-hedging, traits such as pinocytosis or phagocytosis, which are shared by both protozoans and malignant cells (see [20, 21]), are also promising to explore. Similarly, the relationship, if any, between the re-acquisition of motility in cancer cells, and the initial acquisition of motility in apparently archaic eukaryotes [22], could be very helpful in testing the atavistic hypothesis.

Another research direction would consist of exploring whether or not the first mutations leading to carcinogenesis occur or not in the parts of the genome that are involved in natural cancer defense (e.g. tumor suppressor genes). **Natural defenses to cancer indeed evolved together with multicellularity, and are thus posteriori to the unicellular period.** Because of this chronological sequence, the atavistic hypothesis will not be able to explain cancers initiated by mutations altering natural defenses to cancer. Interestingly, mutations on TP53 sometimes occur only in advanced cancers [23], suggesting both that cancer initiation did not always rely on alterations of natural cancer defenses (i.e. in favor of the atavistic model at initiation), but that random mutations can be subsequently favored by selection (in favor of the de novo selection at later stages). More research in this direction for different cancers and for different species would be necessary to clarify this point.

Atavism vs. the somatic mutation theory: Different therapeutic opportunities?

Both strategies discussed here, atavism and bet-hedging, are associated with genomic instability and alteration. For atavism, the genome might be scrambled as a consequence of a reduced DNA repair capacity, and hence unable to meet the needs of a complicated, evolved, high-maintenance genome; alternatively, it might be programmatic, i.e. a form of risk mitigation in the face of a hostile, shifting environment and even include a type of bet-hedging itself. For bet-hedging associated with the somatic mutation theory, however, the genomic instability is believed to result from a random mutator phenotype, but modulated by an evolving bet-hedging strategy.

Either way, the net result on the genome might be remarkably similar. Consequently, therapeutic exploitations of genomic instability (i.e. conventional cytotoxic chemotherapy and radiotherapy, DNA repair-inhibitory drugs and immune checkpoint inhibitors) – all of which have been relatively successful – cannot be used to distinguish between these two theories. Whether one believes in atavism, or the somatic mutation theory, or even some other theory such as aneuploidy, it seems that genomic instability can be exploited as one of the vulnerabilities of the cancer cell, either by increasing the chaos beyond the bounds of viability (chemo or radiotherapy) or by alerting the immune system to the neo-antigen (“non-self”) signal that emerges from the mutational burden. None of this is intrinsically different for any of the underlying theories.

Is therapeutic resistance evidence of atavism?

Resistance to a range of therapies is also characteristic of cancer, and because it is heritable, likely relates to genomic plasticity, which in turn is likely related to the basic mechanisms of genomic instability. As such, it would be consistent with either atavism or somatic mutation, and would not facilitate differentiation between the two. On the other hand, as noted by Vincent [7], there is a remarkable

commonality between today's standard anti-cancer treatments and ancient existential threats (e.g. radiation exposure, natural toxin products, nutritional challenges). Taken together this does suggest an additional pathway to therapeutic resistance, i.e. that eukaryotes had to have “learned” how to survive the pre-Cambrian (i.e. the Proterozoic), otherwise we would not be here today. Consequently, we already “know” how to overcome these modalities when used as therapies, an explanation unique to atavism.

Evidentiary support: “Consistent with” does not equal “predicted by”

It seems that the general opportunities provided by the disordered genome can be predicted (or could have been predicted) from the somatic mutation theory. Insofar as the cancer cell, and tumors as a whole, do indeed exhibit other vulnerabilities, these are also consistent with the somatic mutation theory. These vulnerabilities, and the opportunities that arise from them, will be further elaborated below; but it is important to appreciate that they are not actually predicted from the mutation theory alone, rather they are merely consistent with it.

Importantly there exists a key difference between “prediction” and “consistency”. For instance, whatever the characteristics of cancer are found empirically to be, they are consistent with mutationism; because these characteristics exist, they are assumed to be beneficial, and therefore selected in a Darwinian manner; and to the extent they seem to be novel, they are assumed to be generated by some mutation. Although this logic is not quite circular [24], it is not readily subject to falsification, and hence its scientific status as an explanation is dubious.

Is the somatic theory circular?

Atavism, on the other hand, specifically restricts these apparently **novel characteristics of the cancer cell to a re-embodiment of the features of the ur-karyote, i.e. the earliest eukaryotic cells thought to have initially evolved in the Proterozoic oceans about 1.6 BYA** [25]. Indeed, malignant cells carry several signature features of the ur-karyote phenotype, such as adaptation to hypoxia, Warburg-type metabolism, ROS sensitivity, a substantial pH gradient across the cell membrane, milieu acidification, bloom-like growth, hyperphagia/scarcity adaptation, colonial growth patterns, microcarnivory, asexual reproduction, and immortality. These characteristics not only seem to distinguish cancer cells from normal cells, but resemble the conceptually reconstructed eukaryote, to a greater or lesser degree, given the geochemistry of the Proterozoic [26–31].

Difficulties in testing evolutionary hypotheses

Unfortunately, we do not have a “living fossil” of the ur-karyote, nor of its Proterozoic environment, by which to verify this hypothesis independently; all we have is inference. The difficulties associated with evolutionary explanations have

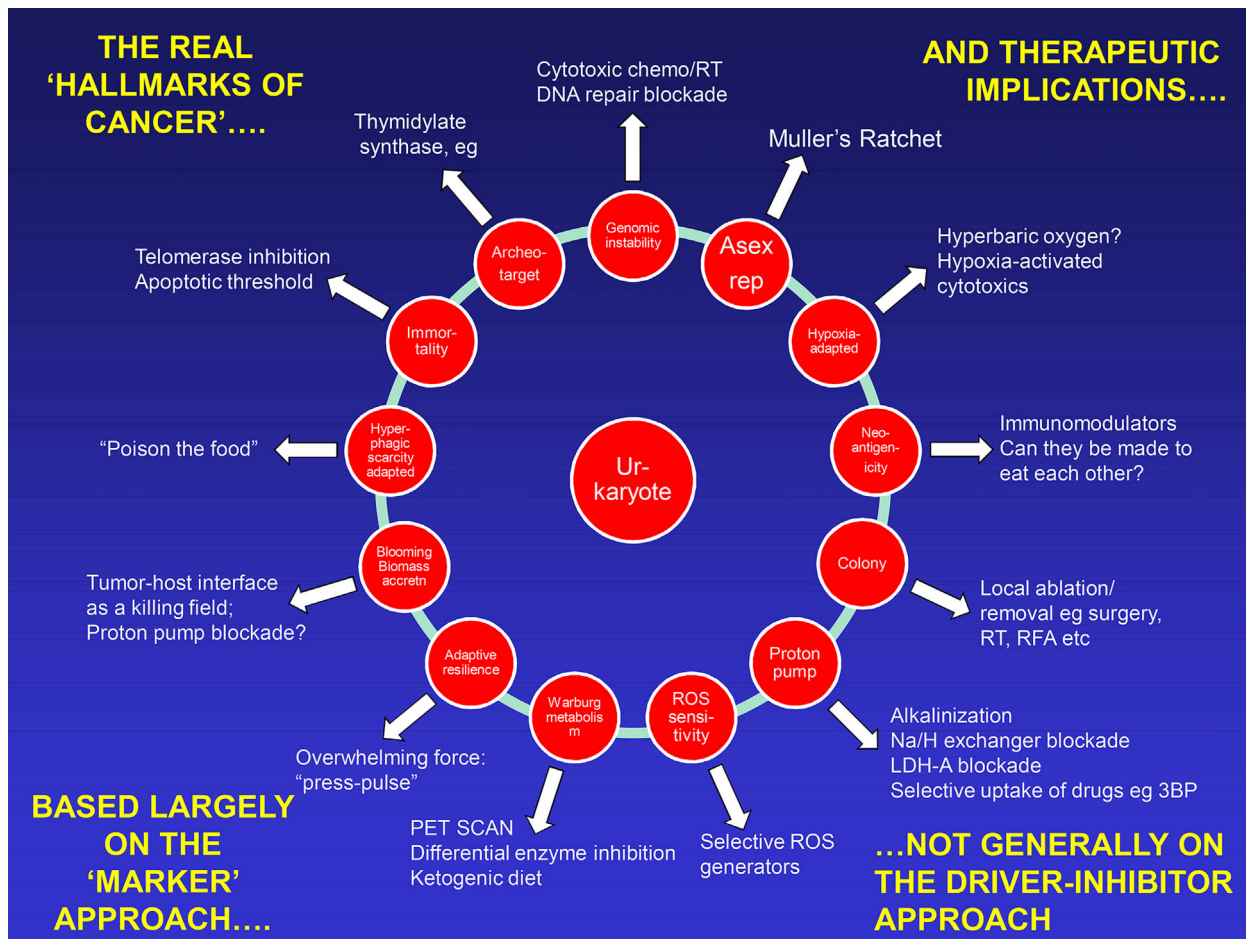


Figure 2. Vulnerabilities of cancer correlated with the signatures of atavism.

been eloquently documented [32] and include "... the interpretation of causes that are almost certainly contingent, multifactorial, interactive, hierarchical, non-linear, emergent, and probabilistic rather than deterministic". Nonetheless, the evolution of life and the ancient environments in which it occurred, are among the most intensely studied of subjects, and it is clear from multiple lines of converging evidence that we do indeed know something rather than nothing. This is why we here offer atavism as a serious candidate hypothesis for the emergence of cancer.

Atavism predicts key vulnerabilities of the cancer cell

The vulnerabilities of cancer, correlating with the signature atavisms of the malignant phenotype, are outlined earlier and in Fig. 2. Note that generally the corresponding therapeutic opportunities that emerge are based on signature recognition and, in general, not causality-reversal. Some of these opportunities have already been discovered empirically (e.g. colonial-type growth resulting in "tumors", which can be locally ablated by a large variety of means, e.g. surgery, etc.) and thus cannot be regarded as "predictions", in the strict sense of the term, but there is a sense in which this could

have been predicted from atavism but not from any other theory. Other potential therapies have yet to be established (e.g. hyperbaric oxygen, ketogenic diet, hypoxia-activated drugs, inhibition of the acid pump, selective ROS generators, telomerase, and other relatively cancer-specific enzyme inhibition suggested by Warburg metabolism) but are under active investigation. Others, such as the PET scan, exploit the presence or absence of the Warburg Effect for accurate staging and diagnosis, and indirectly affect therapy.

Is atavism the unifying hypothesis of cancer biology?

Most of these features were discovered without having any theoretical underpinning; instead their revelation is derived from individual empirical findings. Although consistent with mutationist theories, these features will only "make sense" in the context of atavism. Consequently, atavism offers an underlying explanation of a broad range of ostensibly independent facts, and as a hypothesis, it is potentially falsifiable, i.e. in the case that the characteristics of the malignant phenotype transpire not to be consistent with what we independently know to be the geochemistry of the mid-Proterozoic.

Box 1

Glossary

Bet-hedging: The theory of bet-hedging was developed to explain the presence of ungerminated seeds in the soil (seed-banks) that serve as reservoirs to overcome unexpected environmental challenges, e.g. drought. By producing seeds that germinate at different time points (immediately or laying dormant for an extended period), “hedging their bets” and not putting “all their eggs in one basket” will provide the plants with evolutionary advantage to overcome unexpected environmental perturbances. This strategy ensures that the organism will increase its fitness under stressful conditions, but at the expense of decreased fitness under typical conditions. Other bet-hedging examples include bacterial persistence in the presence of antibiotics, female multiple mating, and nutrient storage in rhizobia.

Ancestral legacy: Characters in different organisms that are similar because they are inherited from a common ancestor that also had that character, also called homologous characters.

Convergent evolution: Morphologically similar traits that have evolved in two or more lineages independently, not from a shared common ancestor (i.e. fins in whales and fish, wings in birds, and bats).

Pinocytosis: Is a form of cellular transport, when extracellular liquids are transported inside the cell via engulfing (invagination) and suspension within small vesicles. Following suspension, the fluid is released inside the cells via hydrolysis of the pinocytotic vesicles.

Phagocytosis: Is a process in which solid particles (e.g. nutrients) are transported inside the cells via engulfing and forming an internal compartment, the phagosome.

Phenotypic plasticity: Organisms are capable of changing their phenotype in response to environmental variation: a fundamental organismal process to cope with environmental variation.

Phylogeny: The study of relationships, including evolutionary history between individuals, species or populations.

Pre-Cambrian period: The dawn of Earth’s history, spanning from ~4.567 billions of years ago (BYA), the formation of Earth, to ~541 million years ago (MYA), the beginning of the Cambrian Period.

Proximate causation: A proximate cause is immediately responsible (i.e. mechanistic precursors) for causing something observed; in contrast, the ultimate cause is considered the underlying or real cause (evolutionary reason).

The prioritization of therapeutic empiricism

To the extent that these signature vulnerabilities actually succeed in being exploited for therapy, especially as a package, they would tend to provide some differential support for atavism rather than a purely random mutator phenotype. Nonetheless, genomic instability is as much a feature of atavism (at least as conceived by one of us [7]) as it is a feature of the somatic mutation theory. Furthermore, the immense value of the therapeutic opportunities emanating from studies of genomic instability (especially immunotherapy, but also DNA repair inhibition) will continue to attract the lion’s share of attention, but without offering a clean way to distinguish between the various theories of cancer.

Conclusions and outlook

Genomic alteration and instability, the *sine qua non* of transformation, are at the very core of cancer biology. It is of paramount importance to understand not only how they are mediated and what they imply for therapy, but also what they mean in a broader sense. While 31 “mutational signatures” have been described in a range of cancers, only some signatures are due to well-defined exogenous mutagens or endogenous repair deficiencies; the origin of many signatures remains obscure [33]. Moreover, genomic disorder in cancer extends far beyond simple point mutations and indels to include large scale aneuploidy and epigenetic derangements; the toleration of this entire package remains one of the most striking features of the

malignant phenotype. We have here suggested two potential origins for the disturbed cancer genome: stochasticity (the somatic mutation theory, i.e. random mutagenesis) and atavism (in which a reduced and simplified DNA repair capacity fails to cope with the complex evolved metazoan genome), and proposed approaches that might serve to distinguish between these hypotheses, while remaining sensitive to the possibility that they are not necessarily mutually exclusive.

A further twist is that this genomic evolvability might subserve a survival function, such that short-term suboptimal fitness from hypervariant progeny is traded off against long-term reproductive success in the face of unpredictable environmental change (“bet-hedging”). As such, the evolvability itself might (have) become entrained as a program, either *de novo* (somatic mutation) or in antiquity (atavism). Either way, bet-hedging is probably responsible for inherent and acquired resistance to therapy, the single most important cause of ongoing cancer mortality today. The disordered and highly heterogenous cancer genome, impossible to repair, can however be approached by strategies to increase the chaos beyond viability, or by immunological exploitation of the contingent “non-self” proteome.

Genomic instability is one of the vulnerabilities of the cancer cell possibly related to atavism, some of which have been discovered empirically without due regard to any unifying theory. Clinical trials of therapeutic interventions which indirectly reflect these opportunities (e.g. DNA repair inhibition, immune checkpoint manipulation) might allow some test of underlying deeper theories, but given the high costs, high risks, and high stakes, will overwhelmingly

continue to focus on short-term empirical success. Inferential support for, or refutation of, evolutionary explanations of cancer will continue to remain a challenge.

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